Tsuboyama epistasis prediction — all double mutants

Original ProteinGym scoring

Conditional masked-marginal

Pseudo-log-likelihood (PLL)

Best-model representative

DeepSequence_ensemble
ESM-IF1
EVE_single
GEMME
PoET
Progen2_xlarge
Progen3_3b
ProtGPT2
ProteinMPNN
RITA_xl
TranceptEVE_M
Tranception_M
Unirep
Wavenet
CARP_76M
ESCOTT
ESMC-300M
ESM1b
ESM1v_ensemble
ESM2_150M
ESM3
MIF
MIFST
MSA_Transformer_ensemble
MULAN_small
ProSST-1024
ProtSSN_k30_h1280
RSALOR
S2F_MSA
S3F
SaProt_35M_AF2
SiteRM
Site_Independent
VESPA
VESPAl
VespaG
VenusREM
xTrimoPGLM-10B-MLM
EVmutation

Spearman ρ: predicted vs experimental thermodynamic coupling

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

AMFR_HUMAN 4G3O
BBC1_YEAST 1TG0
BCHB_CHLTE 2KRU
CATR_CHLRE 2AMI
CBPA2_HUMAN 1O6X
CBX4_HUMAN 2K28
CSN4_MOUSE 1UFM
CUE1_YEAST 2MYX
DNJA1_HUMAN 2LO1
DOCK1_MOUSE 2M0Y
EPHB2_HUMAN 1F0M
FECA_ECOLI 2D1U
HECD1_HUMAN 3DKM
ISDH_STAAW 2LHR
MAFG_MOUSE 1K1V
MBD11_ARATH 6ACV
MYO3_YEAST 2BTT
NKX31_HUMAN 2L9R
NUSA_ECOLI 1WCL
NUSG_MYCTU 2MI6
OBSCN_HUMAN 1V1C
ODP2_GEOSE 1W4G
PIN1_HUMAN 1I6C
PITX2_HUMAN 2L7M
POLG_PESV 2MXD
PR40A_HUMAN 1UZC
PSAE_PICP2 1PSE
RAD_ANTMA 2CJJ
RBP1_HUMAN 2KWH
RCD1_ARATH 5OAO
RCRO_LAMBD 1ORC
RD23A_HUMAN 1IFY
RFAH_ECOLI 2LCL
RL20_AQUAE 1GYZ
RPC1_BP434 1R69
SAV1_MOUSE 2YSB
SDA_BACSU 1PV0
SPA_STAAU 1LP1
SPG2_STRSG 5U8S
SPTN1_CHICK 1TUD
SR43C_ARATH 2N88
SRBS1_HUMAN 2O2W
TCRG1_MOUSE 1E0L
THO1_YEAST 2WQG
TNKS2_HUMAN 5JRT
UBE4B_HUMAN 3L1X
UBR5_HUMAN 1I2T
VILI_CHICK 1YU5
YAIA_ECOLI 2KVT
YNZC_BACSU 2IVD

Bold model names: at least one new scoring method recomputed for all 50 assays. Gray cells: that method is unavailable or still running.